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Microcontainer-Based Self-Healing Cement Additive to Effectively Seal Cracks by Generating Polyurethane Foam

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Abstract

Maintaining long-term well integrity is essential for conducting oil and gas production in an economically and ecologically responsible manner. Here, we developed self-healing characteristics in cement by formulating a complex microcontainer (CMC) additive. The mechanical activation of microcontainers during cracking triggers a self-expanding polyurethane foam production mechanism in the fractured region, which leads to the restoration of the well integrity. The complex microcontainers feature a unique structure that includes the main components of the widely used polyurethane foam, which are enclosed and inert to each other: polyol-containing (component A) polyurea microcapsules within the isocyanate phase (component B) in the core of larger polyurethane microcapsules. We obtained the complex microcontainer system in two steps using interphase polymerization, which followed the creation of oil-in-oil and oil-in-water emulsions. FTIR, SEM, optical microscopy and TGA were used to characterize the obtained material. Additionally, we used model cement samples to investigate the material's ability to foam in the presence of water and its self-healing properties. The foaming characteristics in the aqueous system exhibit effective self-expansion, and 5% wt. of self-healing additive is adequate to successfully seal the fractures within 10 days. The implementation of the proposed self-healing cement can save 10 tons of CO₂ per deployment by minimizing remedial cementing and hydrocarbon leakage.

Introduction

Maintaining well integrity is crucial for ensuring safe, efficient, and environmentally responsible oil and gas operations. A well with compromised integrity can lead to catastrophic failures, including blowouts, leaks, and uncontrolled hydrocarbon releases, potential risks to the environment, and asset value (*Kiran 2017*). Sustaining well integrity involves regular monitoring, proper casing and cementing, and pressure management to prevent sustained casing pressure (SCP) and casing corrosion. Overcoming casing corrosion and annulus (CCA) issues requires advanced materials, corrosion inhibitors, and cathodic protection, along with diagnostic tools. Traditional repair methods, such as cement squeezing or casing patches, are costly and require manual intervention. Self-healing materials are a game-changer for well integrity, offering autonomous repair of cracks and corrosion damage without human intervention.

Usually, self-healing materials rely on embedded microcapsules that rupture upon damage, releasing healing agents to repair cracks. In general, microencapsulation is widely used in a wide variety of industries, for example, to create smart coatings, medical applications and building materials (*Mishra 2019*).

There are a number of self-healing mechanisms for microcapsule-based materials (*Wu 2012*). For example, autonomous repair, when microcapsules filled with monomers, resins, or corrosion inhibitors break upon crack formation, releasing healing agents. The second option for the systems with epoxy or polyurethane-based chemical encapsulation that can harden upon contact with catalysts or moisture. Mineral precipitation also can be used as a healing mechanism: sodium silicate reacts with cement to form calcium silicate hydrate (C-S-H) and seals the crack.

The most common materials of the microcapsules' shells are urea-formaldehyde, melamine-formaldehyde, polyurethane. These materials were not chosen by chance, since microcapsules must withstand aggressive well conditions so that premature activation of the capsules does not occur and the self-healing properties of cement are preserved. As for the fabrication methods, in general, microcapsules can be obtained by interfacial polymerization in emulsion (*Song 2017*), solvent evaporation (*Li 2008*), or layer-by-layer assembly (*Borges 2024*) methods.

It is known that using the above-mentioned materials, one can heal the cracks with width up to 300–500 μm (*Muhammad 2016*). However, unfortunately, for cracks with a larger width, there is still no effective crack-healing method, which is due to the limited swelling/expansion of the material in the crack. The material formed in the crack during healing should effectively fill the crack; therefore, in this case, it would be effective to use a self-expandable filler, for example, polyurethane foam.

Polyurethane foam is a versatile polymer material with a wide range of formulations, offering unique benefits in flexibility, durability, insulation, and more. This material is widely used in industry due to its exceptional insulation properties, high strength-to-weight ratio, long lifespan due to high resistance to wear and moisture (*Ates 2022*). Moreover, it has a fast curing time (can be varied by the presence of a catalyst in the mixture or the temperature).

Polyurethane foam usually consists of two main components, such as polyol (component A) and isocyanate (component B), blowing agents (to increase foam expansion), catalysts (to control reaction speed), surfactants (to stabilize cell structure), and other additives that also can be added to the mixture to enhance the materials properties.

Inspired by this material, we propose a novel complex micro-container (CMC) system that transports a compartmentalized reaction mixture of the common components found in polyurethane foam. The system is composed of polyurethane microcapsules with a core containing two key components: microcapsules filled with polyol (component A) and an isocyanate-based phase (component B). When these capsules break, their contents mix, initiating the healing process. This design allows for effective crack repair even without water, while the presence of water triggers the formation of polyurethane foam, further enhancing healing efficiency. In this work, we employed a two-step interfacial polymerization in emulsion method to fabricate these complex microcontainers. The resulting structures were analyzed using FTIR, SEM, and TGA to verify the dual-shell complex composition and successful cores encapsulation. We also examined the material's foaming behavior and its effectiveness in cement crack repair, utilizing tomography for self-healing assessment. The proposed material is anticipated to improve microcrack sealing efficiency with reduced microcapsule usage while offering better control over the self-healing mechanism.

Description and Application of Equipment and Processes

Materials

Cyclohexanone, cyclohexane, gum arabic, pentaethylenhexamine, and sodium dodecyl sulfate (SDS) were obtained from Sigma-Aldrich. Additionally, oil-in-oil emulsion stabilizers, including polyglyceryl-3-diisostearate, polyglyceryl-3 polyricinoleate, and a styrene/ethylene/propylene copolymer, were utilized

in the study. Toluene-2,4-diisocyanate (TDI), isophorone diisocyanate (IPDI), and Hexamethylene diisocyanate (HDI) were tested as component B - isocyanate. 1,4-butanediol (BDO) and 1,2-propanediol (PDO) were used as component A. These materials also were purchased from Sigma-Aldrich. All materials were used as received without additional purification.

Procedure

As mentioned in the introduction, one common microencapsulation technique is based on interfacial polymerization in an emulsion. First, an emulsion of two immiscible liquids is created. The interfacial polymerization process then occurs in two steps. During the initial stage, two reactive monomers diffuse and meet at the interface—specifically, the surface of the oil droplets in the emulsion. This interaction results in the creation of a thin primary polymer shell. In the subsequent step, the shell thickens due to continued polymerization reactions. As a result of the polymerization process, microspheres containing some core materials are formed (*Kardar, 2015*).

In this work several encapsulation technologies (*Lin 2022, Li 2022, Kobaslija 2006, Yang 2008*) were adjusted to create the complex micro-containers and the synthesis procedure was carried out in multiple steps:

1. Preparation of an oil-in-oil emulsion;
2. The polyurea shell formation via interfacial polymerization;
3. Preparation of an oil-in-water emulsion containing the synthesized microcapsules;
4. The polyurethane shell formation via interfacial polymerization.

Following the careful selection of materials and synthesis parameters, TDI was chosen for creating a polyurea capsule shell, while IPDI was chosen for the core material of the large polyurethane microcapsules. The synthesis of microcapsules with different polyols did not show any significant changes in capsule morphology, however, this paper presents the results for the CMCs with BDO. More synthesis details can be found in the [Figure 1](#) and in the works (*Stamer 2025, Baltaeva 2023*).

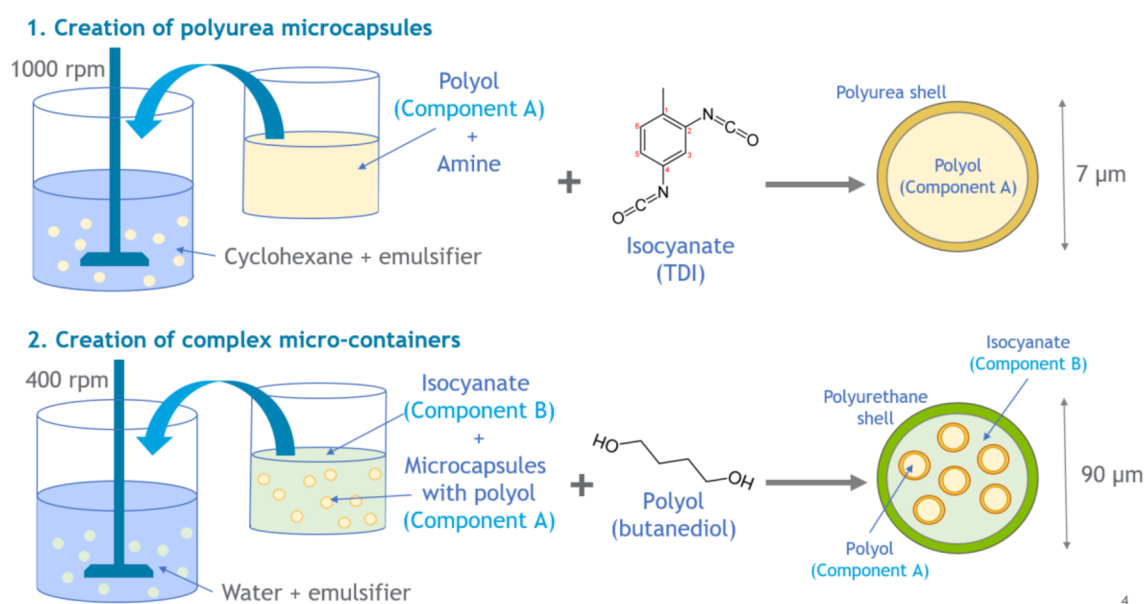


Figure 1—General mechanism of the CMCs preparation.

The composition, structure and properties of synthesized materials were verified using SEM, TGA, FTIR.

Polyurethane foam formation

To demonstrate the self-expanding properties of the synthesized material, we conducted a model test. First, 2 g of crushed CMCs were placed in a conical tube at room temperature. Then, 40% wt. of a 1% aqueous SDS solution was added to the tube, and the mixture was gently homogenized using a spatula. The initial volume of the mixture was recorded. The surfactant (SDS) was incorporated into the system to stabilize the cellular structure. The conical tube, secured in a stand, was then incubated in a heating chamber at 60°C for 2 hours. After this period, the final expanded volume of the mixture was measured to assess the expansion behavior.

Cement samples preparation and self-healing process evaluation

For testing CMCs' performance, a standard cement slurry formulation with water / (cement (+CMCs)) ratio = 0.45, without additives that can affect the results of experiments, with known setting time, rheological parameters and compressive and bending strengths, was taken.

For the preparation of cement mortar 5% wt. of the cement mass were replaced by CMCs. To prevent damage to the microcapsules, they were carefully blended with dry cement in a beaker using a spatula. Deionized water was then added, and the mixture was stirred to ensure proper workability. The fresh cement paste was poured into molds to produce 40×40×40 mm cubes, which were demolded after curing under conditions tailored to the cement's initial composition.

Cracks were induced using a hydraulic press, applying a load equivalent to 70% of the bending strength limit. The required load was determined through preliminary tests on control cement samples without microcapsules (CMCs). The self-healing process was conducted in a water bath maintained at 25°C.

The healing performance of the CMCs was visually assessed using micro-computed tomography (μ -CT) after specified time intervals. This non-destructive technique provides high-resolution 3D imaging by capturing multiple X-ray projections of the sample and reconstructing them into a volumetric model. The system is particularly suited for detailed geometric analysis of small- to medium-sized specimens.

Data and Results

Methodology development

During the synthesis of CMCs, it was found that the morphology and size of microcapsules greatly depend on a variety of factors, including dispersion speed, temperature, polyol/isocyanate ratio (for the synthesis of the outer shell), amount of the smaller microcapsules added, etc.

The synthesis of microcapsules was quite complicated due to several reasons. We varied some of the key parameters mentioned above to see how they affect synthesis. As a result, it was found that the key factor ensuring the stability of the emulsion (the oil-in-water emulsion) is the amount of polyurea microcapsules added to the dispersed phase. The more of them there were in the emulsion droplets, the less stable the system was. Moreover, it was determined that increased loading of polyurea microcapsules resulted in the low loading of the isocyanate in CMCs and in ineffective foam formation after CMCs breakage.

Based on these results, we found the optimal ratio corresponding to the maximum possible loading of microcapsules into the outer shell. This requirement is necessary to ensure more effective crack healing. The more microcapsules in the polyurea shell, the more likely they will be damaged and release polyol, which must react with isocyanate.

For more information on the materials screening, as well as experimental detail and adjustment of the synthesis parameters, please refer to our previously published works (*Stamer 2025*, *Balateva 2023*).

Morphology

The microcapsule morphology was analyzed using optical and scanning electron microscopy (SEM), as shown in [Figure 2](#). Both techniques verified the successful encapsulation of smaller polyurea microcapsules within larger polyurethane ones, consistent with the expected structure. SEM examination revealed that

the inner polyurea microcapsules (containing polyol) ranged in size from a few micrometers, while the surrounding polyurethane microcapsules measured approximately one hundred micrometers in diameter. More information regarding synthesized materials' morphologies can be found in the work ([Stamer 2025](#)).

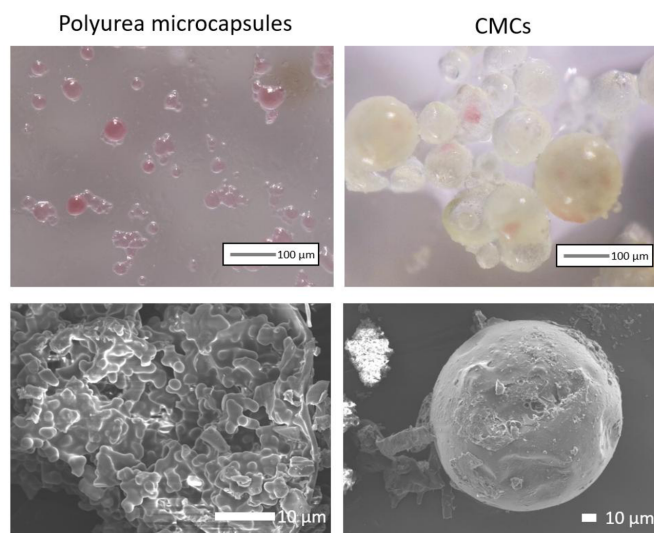


Figure 2—Visual (upper) and scanning electron microscopy (lower) micrographs of the synthesized microcapsules.

FTIR and TGA

FTIR analysis was employed to characterize the chemical composition of the microcapsules by identifying functional groups, confirming the presence of specific compounds in both the shell and core materials.

The FTIR spectra in [Figure 3](#) reveal characteristic absorption bands confirming the microcapsule composition. A strong peak at 2258 cm^{-1} , corresponding to $\text{N}=\text{C}=\text{O}$ stretching vibrations, indicates unreacted isocyanate ($-\text{NCO}$) groups in the core, verifying successful encapsulation. Distinct amine group peaks appear at 3326 cm^{-1} (CMC) and 3288 cm^{-1} (polyurea) for N-H stretching, along with bending vibrations at 1531 and 1539 cm^{-1} . These features confirm amine group presence in both polyurethane outer and polyurea inner microcapsule shells. is related to the polyurethane and polyurea shell formation respectively. The carbonyl group shoulder observed at around 1700 cm^{-1} ($-\text{C}=\text{O}$ from urethane) confirms the presence of urethane linkages.

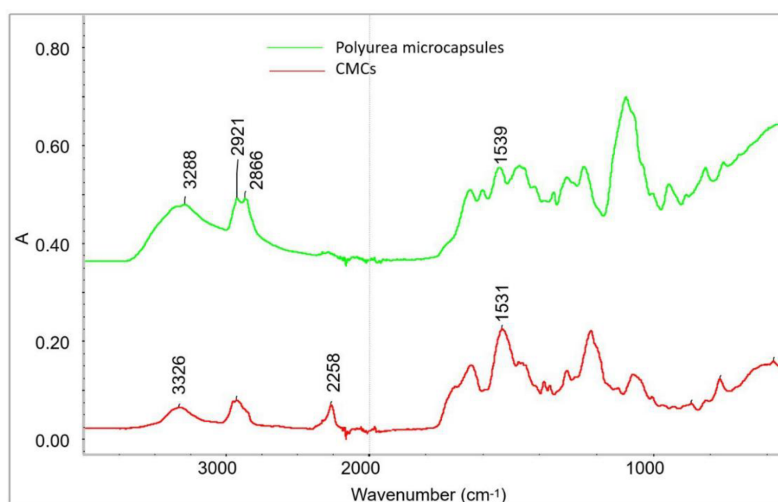


Figure 3—The ATR-FTIR spectrum of microcapsules.

From the spectrum of polyurea microcapsules it is possible to observe the characteristic symmetric and asymmetric stretching vibration of $-\text{CH}_2$ bond occurring at 2866 and 2921 cm^{-1} , respectively, which indicates polyol content in polyurea microcapsules. Also, one can see the overlapping of the $-\text{OH}$ stretching vibrations peak, related to polyol, and $-\text{N}-\text{H}$ stretching vibrations peak, related to the presence of amine groups, at 3288 cm^{-1} .

TGA was carried out to measure the mass loss of the microcapsules along with a certain range of temperature under a nitrogen atmosphere with a controlled heating rate of $10\text{ }^{\circ}\text{C}$ per minute. For both types of microcapsules, inner and outer, the decomposition of microcapsules consists of three stages:

1. Residual solvent evaporation;
2. The decomposition of encapsulated material;
3. The shell undergoes two-stage thermal decomposition: initial degradation of soft segments derived from active hydrogen sources and chain extenders, followed by decomposition of hard segments.

Figure 4 presents the results of TGA measurements. The significant weight loss observed between 100 – 200°C corresponds to decomposition of the polyurea microcapsules' core. The shell degradation initiates at approximately 200°C and proceeds through two distinct stages: initial breakdown of soft segments (200 – 400°C), followed by decomposition of hard polymeric segments above 400°C .

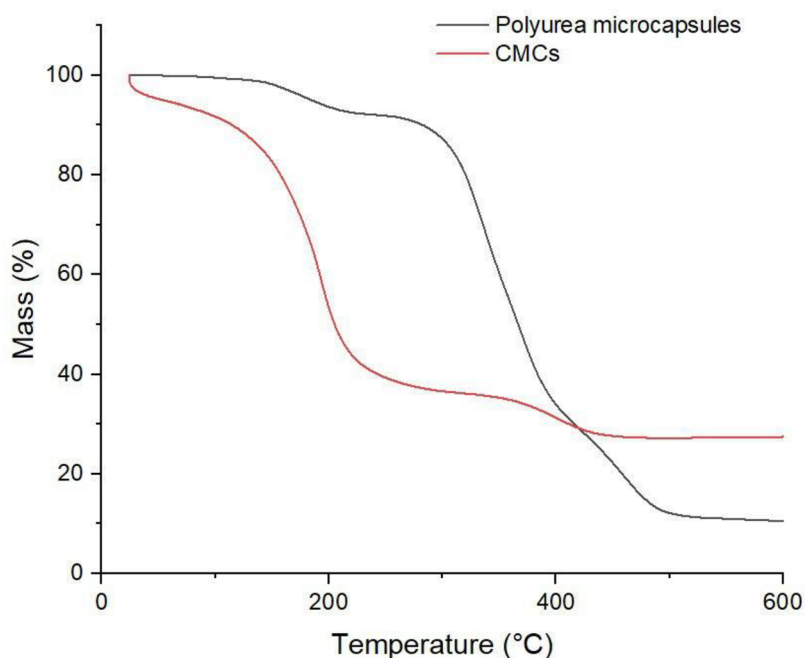


Figure 4—TGA weight loss curve of synthesized polyurea microcapsules.

The CMCs' complex structure makes it hard to tell which microcapsule material components are degrading. However, it is easy to see that CMC is much more stable at high temperatures due to the presence of an outer shell (degradation of the CMCs' core material is not observed up to almost 180°C , which is more than sufficient for oil and gas production applications).

Foaming properties and testing in cement

It is known that polyurethane foam is formed through a reaction between polyols and diisocyanates, with additional additives controlling foam structure, density, and properties. The foam synthesis follows the step-growth polymerization mechanism, in which polyol reacts with diisocyanate to form polyurethane material.

In parallel, a foaming reaction with CO₂ gas release occurs while water comes into contact with isocyanate. CO₂ gas creates the foam's cellular structure.

The photo in [Figure 5](#) shows only an example of the foam formed after CMCs rupture in the presence of water. A more detailed study of foaming depending on the amount of water in the system and temperature, carried out by us, is given in the reference. The photo shows the porous material, which has doubled in volume. Such self-expansion can ensure the healing of large-diameter cracks. It is evident that the quality of self-expansion will depend on a number of parameters, such as the temperature inside the well, the amount of water, and so on. It is also worth noting that the size of the pores formed during the process has the potential for both enhancing and impairing effects on resulting cement's mechanical strength. At the same time, it is worth noting that the pressure in the well will also have a negative effect on the effectiveness of self-expansion of the synthesized foam. We should further investigate these issues before conducting materials field testing.



Figure 5—Example of foaming properties.

The FTIR spectroscopy of the obtained polyurethane foam is shown in [Fig. 6](#). In the presented spectrum one can see N-H stretching at 3334 cm⁻¹, C=O stretching at 1635cm⁻¹, and N-H bending and C-N stretching at 1546cm⁻¹ that are related to the urethane linkage ([Jiao 2013](#)). At the same time there are no C-H stretching vibrations at 2850–2980 cm⁻¹, as well as N=C=O stretching. These mean that there is no unreacted isocyanate or polyol in the system.

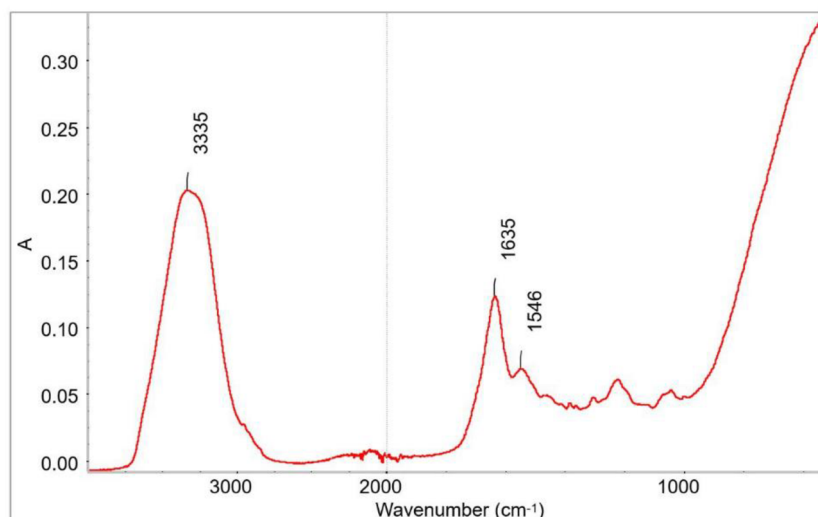


Figure 6—FTIR spectrum of the polyurethane foam.

The healing performance evaluation data for the CMCs is presented in Fig. 7. The presented 3D-volume representation of the samples shows the effectiveness of healing cracks in the cement cube. It can be seen that after 10 days of exposure at room temperature, both the number of cracks and their thickness decrease. Under elevated temperatures typical of well conditions, or when incorporating a catalyst into the capsule system, both the crack healing rate and material expansion efficiency are enhanced. This synergistic effect significantly improves the overall self-healing performance. More information regarding materials' self-healing ability, as well as cement strength restoration, can be found in the work (*Stamer 2025*).

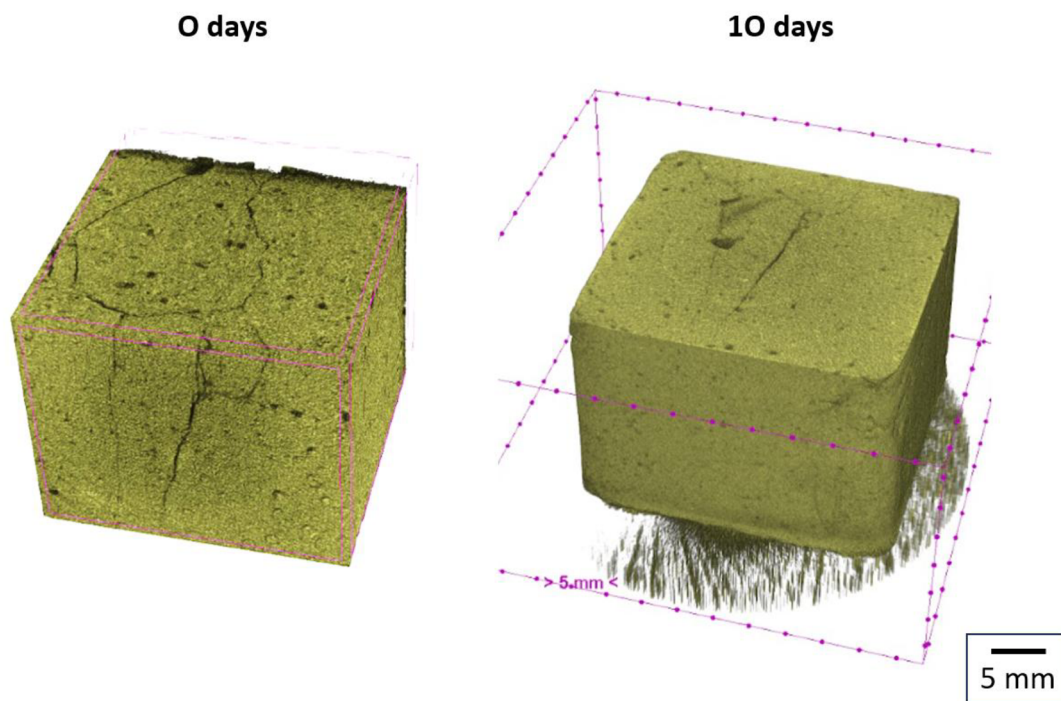


Figure 7—Examples of self-healing process in test cement cube.

Future perspectives

Despite their promising potential, microcapsules for self-healing cement face several key challenges. First, encapsulation efficiency is critical because the shell material must protect the healing agent (e.g., polymers,

minerals, or bacteria) during cement mixing without rupturing prematurely. At the same time, it must also break when cracks form. Compatibility issues arise between the microcapsule shell, which is often polymeric, and the highly alkaline cement environment, as the latter can degrade the former over time. The interfacial bonding between the shell material and cement matrix represents another critical performance parameter requiring evaluation. Another hurdle is achieving uniform dispersion, as microcapsules must be distributed evenly in the cement matrix to ensure consistent healing; however, their low density often leads to segregation. Finally, the long-term durability of microcapsules must be proven, as repeated healing cycles and exposure to harsh conditions may degrade their performance.

Regarding the upscaling of this technology, it is worth noting that, while scaling the synthesis of microcapsules production in an emulsion is feasible, stabilizing the emulsion during creation of the outer shell may present a separate challenge due to the presence of polyurea microcapsules in the system. Therefore, scaling up this technology poses a potential challenge.

Another important aspect is that the pores forming in the system should not exceed a critical size, as this might have a negative effect on mechanical strength. Therefore, a study of the effect of pore size on cement properties should be conducted and conclusions about the need for size control should be formulated.

An interesting area for further research could be the development of a system that produces a more environmentally friendly version of classic polyurethane foam, using, for example, castor oil as a polyol.

Overcoming these challenges requires advances in materials science, encapsulation techniques, and cost-effective manufacturing processes to make microcapsule-based, self-healing cement a viable option for real-world infrastructure projects.

Conclusions

This study provided the encapsulation of a reactive mixture compartmentalized by shells. This synthesized system consists of polyurea microcapsules containing polyol, which are enclosed within a larger polyurethane microcapsule that contains an isocyanate-based phase. FTIR analysis confirmed the successful formation of both polyurea and polyurethane shells, along with isocyanate encapsulation within the polyurethane capsules. SEM imaging revealed polyurethane microcapsules measuring approximately 100 μm , with inner capsules ranging from several micrometers in size. As anticipated, the CMCs demonstrated effective polyurethane foam formation. Computer tomography illustrated the visual, effective crack healing and showed effective crack sealing in the test cube with 5% wt. CMC loading after 10 days of exposure in a water bath. This novel material demonstrates potential to address critical operational challenges in cementing, including long-term zonal isolation failures and casing/casing annulus (CCA) pressure buildup - both of which have significant business consequences. While CMCs show strong promise for well integrity applications, further research is needed to address remaining technical considerations.

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